SYSTEM DESIGN

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1. INPUT DESIGN

The input design is the link between the computers and the users. It composes developing specification and procedure for data preparation and those steps that are necessary to put transaction data into a usable form for processing data entry. Input design is one of the most expensive phase of the development of a system. A large number of problem with a system can usually be raised due to the fault input design and method. Therefore needed to say, the input data is the life block of a system and has to be analysed and designed with the most consideration.

- The decision made during input design must help.
- To provide cost effective method of input.
- To achieve highest level of accuracy.

System analysis decide the following input design details like, what data item to input, how the data should be arranged, coded data items and transactions needing validations to detect errors and finally dialogue to guide the users in providing input. The design of input involves identifying the data needed, specifying the characteristic of each data item, capturing and for computer processing and ensuring correctness of data.

2. OUTPUT DESIGN

The output design refers to the result and information that are generated by the system. For many users output is the main reason for developing a system and the basis on which they evaluate the usefulness of an application.

The objective of a system leads to the determination of output. The analysis of a system leads to the determination of output. Output of a system can take various forms. The most common reports are, screen displays, printed forms, graphical forms etc. the output also varies in terms of their contents, frequency, timing and format. The users of the output, its purpose and sequence of details to be printed are all considered. The output form a system is the justification for its existence. If the output is inadequate in anyway, the system itself in inadequate. The basics requirements of the output are that it should be accurate, timely and appropriate, in terms of content, medium and layout for its intended purpose.

DATAFLOW DIAGRAM

A data flow diagram (DFD)is graphical representation of the “flow” of data through an information system. DFD can also be used for the visualization of the data processing (structured design).A DFD provides no information about the timing or ordering of processes, or about whether processes will operate in sequence or in parallel. Data- flow diagrams can be used to provide the end user with physical idea of where the data they input ultimately has an effect upon the structure of the whole system from order to dispatch to record. How many systems are developed can be determined through a data flow diagram. The circle or bubble represents a transformation process and the label inside the bubble describes the process, using an active verb. Data flows are directed lines that identify the input dataflow and output data flows at each process bubble. Data storage is represented by an open-ended rectangle with a label that identifies the data store or file. The square is labelled to identify an external entity that is a source or destination of a data flow.

Data flow diagrams are used to define the flow of the system and its resources such as information. DFD are way of expressing system requirements in a graphical manner. DFD represents one of the most ingenious tools used for the structured analysis. It has the purpose of clarifying system requirements and identifying major transformation that will become programs in the system design. It is the major point in the design phase that functionality decomposes the requirements specification down the lowest level of details.

Flow diagrams in general are usually designed using designed using simple symbols such as a rectangle, an oval or a circle depicting a processes, data stored or an external entity, and arrows are generally used to

depicts the data flow one step to another. Data flow diagrams present the logical flow of information through a system in graphical or pictorial form.

Basic Dataflow Diagram Symbols

- **Dataflow**

Used to represent functions or process Packets of data to travel from one point to another. Data may flow from a source to process and from data store or process. An arrow line depicts the flow, with arrow head pointing in the direction of the flow. Arrowhead Indicating the direction in which the data is flowing. Can have two arrow heads when a process is altering existing records in a data stores.

- **Process**

Circle stands for processes that convert data into information. A process represents transformation where incoming data flows. A data flow is a route, which enables are changed into outgoing data flows. Must have at least one input and at least one output. When naming process, avoid glossing over them, without really understanding their roles.

- **Data store**

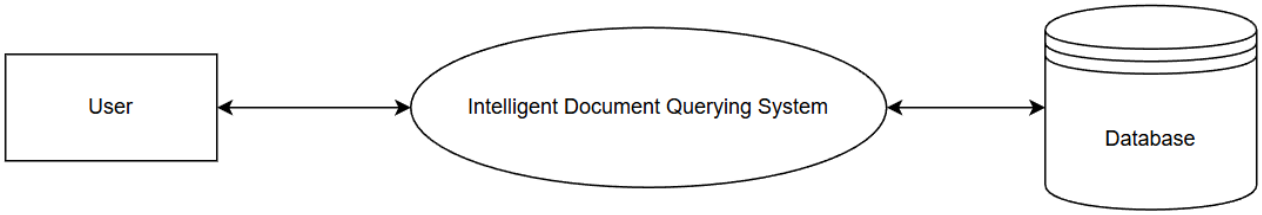
A data store is a repository of data that is to be stored for use by a one or more process may be as simple as buffer or queue or sophisticated as relational database. Tjey should have clear name. If a process merely uses the content of store and does not alter it.

- **Source/Destination**

A source or sink is a person or part of an organization which enters or receives information from the system, but is considered to be outside the content of data flow model. It is normal for all the information represented with a system to have been obtained from or to be passed onto an external source or recipient. These external entities may be duplicated on a diagram, to avoid crossing dataflow lines, they are duplicated a stripe is drawn across the left hand corner.

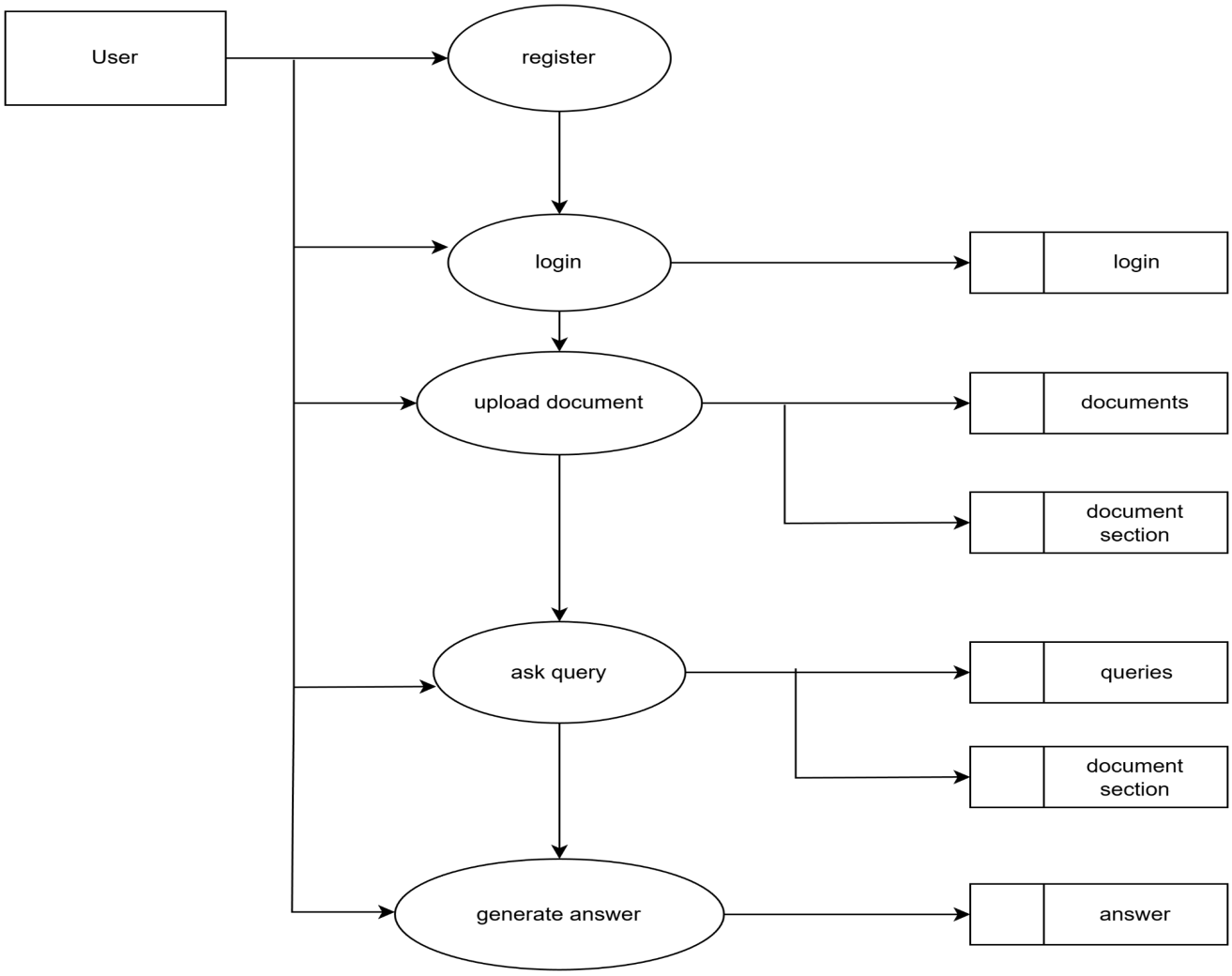
CONTEXT LEVEL

(LEVEL 0):



(LEVEL 1):

LEVEL-1 USER



ENTITY RELATIONSHIP DIAGRAM

An **ER Diagram (Entity–Relationship Diagram)** is a conceptual modeling technique used in database design to represent the structure of a system by showing its entities, attributes, and the relationships among them. It provides a clear visual overview of what data will be stored in the database, how different data objects are related, and what properties each object possesses. ER diagrams act as a blueprint before converting the design into database tables, helping developers and students understand data requirements, reduce design errors, and ensure efficient database organization.

Basic ER Diagram Symbols

- **Entity**

An **entity** represents a real-world object or concept that can be uniquely identified and stored in a database. It may be a person, place, object, or event such as User, Student, or Document. In an ER diagram, an entity is shown using a **rectangle** and usually becomes a table in the database.

- **Attribute**

An **attribute** describes the characteristics or properties of an entity. Examples include Name, ID, Email, or FileName. Attributes are represented using **ovals** and are connected to their respective entities. They define the details that need to be stored for each entity.

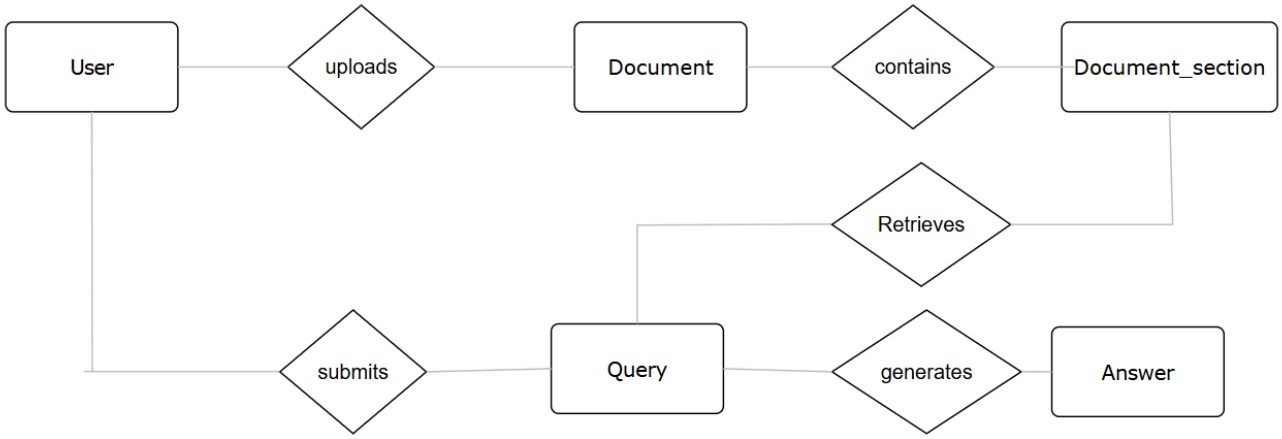
- **Relationship**

A **relationship** shows how two or more entities are connected to each other in the system. It explains the interaction between entities, such as a User *uploads* a Document or a Student *enrolls in* a Course. Relationships are represented using **diamonds**.

- **Connecting Lines**

Connecting lines are used to link entities with their attributes and relationships in an ER diagram. They indicate how elements are associated and may also show **cardinality**, such as one-to-one, one-to-many, or many-to-many relationships.

ER DIAGRAM



DATABASE DESIGN

Database design is one of the most important parts of the system design phase. In a database environment, common data are available and are used by several users. Instead of each program managing its own data, authorized users share data across the application with the database software managing the data as an entity. The primary objective of database design is fast response time to enquiries, more information at low cost, control of redundancy, clarity and ease of use, date and program independence, accuracy and integrity of the system, fast recovery and availability of powerful end-user languages. The theme behind a database is to handle information as an integrated whole, thus the main objective is to make the information as to access easy, quick, inexpensive and flexible for the users. Data directory specifies the major element in the system, and care should be taken while designing, in order to avoid unnecessary duplication of data. The entire package depends on, how the data are maintained in the system. Several tables are maintained in the system to store data that are required for the processing of various data as well as storing intermediate or final processed results. Database design mainly aims at handling large volumes of information, involving the definitions for the structure of storage and provisions for the manipulation of information, providing safety of information despite system crashes due to unauthorized access.

Some conditions are satisfied in the database design stage:

- Control redundancy.
- Ease of use.
- Data independence.
- Accuracy and integrity.
- Recovery from failures.
- Security and privacy.
- Performance

NORMALIZATION

It is a process of efficiency organizing data in database. Normalization has two main goals. One is to eliminating redundant data, ensuring data dependencies make sense. They reduce the amount of space a database consumes, and ensure that is logically stored.

First normal form: First normal form eliminates duplicate columns from the same table. It creates separate table for each group of related data and identify each row with a unique column or set of columns (primary key).

Second normal form: Second normal form meets all the requirements of the first normal form. It removes subset of data that apply to multiple rows of a table and place them in separate tables. Create relationship between these new tables and their predecessors through the use of foreign keys.

Third normal form: Third normal form meets all the requirements of the second normal form. It removes columns that are not dependent upon the primary key.

TABLE STRUCTURES:

Table 1: USER

Description: Data about the user

Field Name	Data type	Description
user_id	int(11)	Primary Key or Not
username	varchar(100)	Username for login
email	varchar(100)	User email
password	varchar(100)	Password for login

Table 2 : DOCUMENT

DESCRIPTION : Data about the document

Field Name	Data type	Description
document_id	int(11)	Unique document
user_id	int(11)	References user
document_name	varchar(150)	File name
upload_date	date	Upload date
file_type	varchar(20)	PDF/ TXT

Table 3 : DOCUMENT_SECTION

DESCRIPTION : Data about the document sections

Field Name	Data type	Description
section_id	int(11)	Section ID
document_id	int(11)	References Document
section_text	text	Document content chunk

Table 4 : QUERY

DESCRIPTION : Data about the user query

Field Name	Data type	Description
query_id	int(11)	Query ID
usert_id	int(11)	References User
query_text	text	User question
query_date	date	Query submitted date

Table 5 : RETRIEVAL_RESULT

DESCRIPTION : Data about the retrieval result

Field Name	Data type	Description
result_id	int(11)	Result ID
query_id	int(11)	References Query
section_id	int(11)	References Document Section
similarity_score	float	Relevance score

Table 6 : ANSWER

DESCRIPTION : Data about the answer

Field Name	Data type	Description
answer_id	int(11)	Answer ID
queryt_id	int(11)	References Query
answer_text	text	Generated answer